

How life gets started, watched inside a computer

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One page, for anyone who wants the idea before the evidence. Everything here comes from experiments we ran; where we say what it might mean for real chemistry, we say so.

What did we find that nobody knew before?

Take twenty thousand tiny computer programs made of random bytes, lay them out on a grid, and let neighbours run each other's code, over and over, with no goal and no scoring. Within a few hundred rounds the same thing happens in almost every world: a program appears that copies itself. It is two bytes long, and it copies itself by accident of being executed: the instruction's own bytes are what it writes. It has no loop and no idea where it ends, so the processor runs straight out of it into the neighbouring program and keeps going. Because of that, what its children look like is mostly decided by the neighbour, not by it: seven of the eight bits that describe a child belong to the neighbour. The first living thing in these worlds is not an individual. It is a colony that works best among its own copies.

Then, in world after world, evolution does the same thing next. A descendant appears carrying one extra instruction, a jump that sends execution back into its own code. From then on the processor never leaves the organism, every neighbour becomes an exact copy, and the children carry zero bits from the neighbourhood. We proved why this had to happen: anything that writes fewer bytes than it reads cannot finish a copy of itself in one pass, so it must either run on into its neighbour or come back on itself. There is no third way.

How does this change the way we think about the origin of life?

Three ways. First, the first living thing was simple, not complex. In our worlds it is the simplest object of its size that exists; by a standard complexity measure it scores the same as the sterile junk around it and lower than the random programs it replaced. If you look for life by looking for complexity, you look in the wrong place at the moment that matters most. Second, being an individual is a later invention than reproducing. Life began open and dependent, and independence had to evolve. Third, independence is not a wall or a membrane. It is a property of control, of where the process goes, and it can be measured as a number of bits that falls to zero.

Why is this significant?

Origin-of-life theories agree that at some point a pattern that merely spread became a thing with a self, whose future depended on itself rather than on its surroundings. Nobody had been able to watch that step, repeat it, and take it apart. Now there is a place where it happens thousands of times with the dials exposed, and the dials are few: whether the chemistry has a channel in which content writes itself decides whether life begins open; whether the sterile junk that forms first is harmless or deadly decides whether the open phase ends in independence, in extinction, or never.

What does the insight let an experimentalist do differently?

If you work on template chemistry: base pairing is such a channel, where a strand writes its own complement with no interpreter. Expect the first replicators to be the shortest self-templating repeats, dependent on their surroundings, making mostly partial copies. So stop hunting for a clean self-replicating species and measure heritable propagation among sloppy copies instead: move the products into fresh material and ask whether the products of the products carry the pattern. Measure openness directly, as how much the product varies when you vary the context. Treat by-products as the variable that matters: screen chemistries for whether their tar stops the copying process, because that, not yield, decides whether an open phase lasts long enough for control to evolve. And look at edges: open replicators should persist in patches of their own copies and individuality should appear where patches meet.

If you study evolution: the first innovations after replication should be about control, where copying starts, stops and returns, not about new functions or codes; and true lineages should appear exactly when one copying event spans the whole genome. If you search for life elsewhere: expect abundant simple order before and around the first replicators, expect every detector that reads structure to fire on it, and design detection as intervention, adding fresh substrate and looking for propagation, rather than as inspection.

What new questions does it pose?

Does the open beginning survive only because neighbours on the grid are its own copies? What happens when the junk kills? Can a chemistry close by copying everything it touches, without a loop, as one of our machines does? Which beginnings are inevitable: our answer is those with a self-writing channel, harmless by-products and time, and the first thing they make is a dependent colony that later learns to stop listening.